

CAP-FL: A Context-Aware Personalized Federated Learning Framework for Next-Word Prediction

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Abstract—Next Word Prediction (NWP) systems are widely used in smart text interfaces, but traditional centralized training puts user privacy at risk and fails to capture diverse language styles. *Federated Learning (FL)* [1] offers a privacy-preserving alternative by enabling collaborative model training across clients without sharing raw data. However, standard FL can perform poorly with *Non-IID* client data [2].

This work introduces Context-Aware Personalized Federated Learning (CAP-FL), a framework for personalized next-word prediction using *DistilGPT-2* in federated settings. CAP-FL employs two key approaches: *Explicit Feature-Based Clustering*, which groups clients with similar text patterns, and *Context Token Injection (CTI)*, which injects structured metadata tokens (e.g., play, genre, scene, character) into input sequences. This two-level personalization enables the model to capture both broad context (genre) and fine-grained context (scene/character), improving coherence and prediction accuracy.

Experiments on the *Shakespeare Non-IID* dataset [19], [20] show that CAP-FL achieves lower cross-entropy loss, reduced perplexity, and higher prediction accuracy compared to standard FL methods such as *FedAvg* and *FedProx* [1], [2]. The framework provides a scalable, interpretable, and privacy-friendly solution for personalized text generation in federated environments. Future work will focus on efficient parameter tuning and real-time context adaptation.

Index Terms—Federated Learning, Next Word Prediction, Context-Aware Personalization, Non-IID Data, DistilGPT-2, Context Token Injection, Clustered Aggregation.

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I. INTRODUCTION

A. Motivation and Background

Next Word Prediction (NWP) represents a cornerstone application of modern Language Models (LMs), enabling intelligent text assistance in communication platforms and productivity tools such as Google Keyboard (Gboard) [1]. These systems rely on accurate, context-sensitive predictions that dynamically adapt to a user's linguistic style. However,

centralized training of such models on user-generated data raises serious privacy and scalability concerns.

Federated Learning (FL) [2], [3] has emerged as a viable paradigm for privacy-preserving model training, where client devices collaboratively train a shared model without exposing raw data. In this setting, local updates are computed on-device and aggregated by a central server to improve a global model. Such decentralized training is particularly suited for next-word prediction, where linguistic diversity across users naturally mirrors the heterogeneity observed in real-world communication data.

B. The Challenge of Non-IID Data in Federated Next Word Prediction

A major limitation of traditional FL lies in its assumption of Independent and Identically Distributed (IID) client data [4]. In practice, data across devices are inherently Non-IID—users differ in vocabulary, syntax, stylistic expression, and contextual focus [5]. This statistical heterogeneity leads to client drift, where local model updates diverge significantly from the global objective, degrading overall convergence and generalization performance [6].

In language modeling, such variation is especially pronounced: each user—or in this study, each Shakespearean play—exhibits distinct linguistic structures and semantic patterns. Consequently, a single global model cannot effectively capture the stylistic and contextual nuances across all clients. This motivates the development of Personalized Federated Learning (pFL) frameworks [7], [8], which aim to tailor global knowledge to client-specific data distributions while preserving privacy.

C. Proposed Solution: Context-Aware Personalized Federated Learning (CAP-FL)

To address these challenges, we propose **Context-Aware Personalized Federated Learning (CAP-FL)**, a framework designed to enhance contextual understanding and personalization in federated next-word prediction using GPT-2.

CAP-FL introduces two complementary mechanisms:

Explicit Feature-Based Clustering: Clients are grouped a priori based on accessible, non-private metadata—in this study, the genre of each Shakespearean play (e.g., Tragedy, Comedy, History). Each cluster maintains its own shared GPT-2 model, allowing aggregation among stylistically similar clients. This approach mitigates Non-IID effects by improving intra-cluster homogeneity and convergence stability [9], [10].

Context Token Injection (CTI): During preprocessing, structured metadata tokens such as <PLAY>, <GENRE>, <ACT>, <SCENE>, and <CHARACTER> are prepended to each input sequence. These tokens serve as explicit conditioning signals, allowing the GPT-2 model to learn semantic and stylistic dependencies beyond standard word co-occurrence patterns [11]. By providing both macro-context (play, genre) and micro-context (scene, character), CTI facilitates fine-grained personalization and improves linguistic coherence.

The current implementation performs full fine-tuning of GPT-2 within a federated setup using the Flower framework. Future work will explore Parameter-Efficient Fine-Tuning (PEFT) methods such as LoRA to further reduce communication and computation costs.

D. Contributions

The main contributions of this work are summarized as follows:

- **Algorithmic Framework:** Introduction of CAP-FL, a dual-layer personalization framework that combines genre-based client clustering with context token injection for next-word prediction using GPT-2.
- **Context-Aware Data Representation:** Design of a pre-processing pipeline that constructs structured input-target pairs with contextual metadata to enable personalized generation.
- **Federated Personalization:** Implementation of a cluster-wise aggregation strategy using the Flower FL framework, followed by local fine-tuning for client-specific adaptation.
- **Empirical Validation:** Experiments on the Shakespeare Non-IID dataset show that CAP-FL achieves faster convergence and lower perplexity than standard FL baselines such as FedAvg and FedProx.
- **Scalability Outlook:** The framework establishes a foundation for future integration of LoRA-based PEFT to optimize efficiency on resource-constrained edge devices.

II. RELATED WORK

A. Personalized Federated Learning (pFL) Taxonomy

Personalized Federated Learning (pFL) techniques aim to address performance degradation caused by heterogeneous client data by allowing each client to retain individuality while contributing to a global model. Conventional pFL methods generally fall into two categories:

Model Regularization Approaches: Methods like FedProx and pFedMe introduce proximal or regularization terms to stabilize optimization across diverse client data distributions [2], [7].

Layer-Specific Personalization: Approaches such as FedPer and Ditto divide the model into globally shared base layers and locally personalized head layers.

While these methods have proven effective for smaller neural architectures, their application to autoregressive language models like GPT-2 remains challenging due to computational complexity and data variability [8], [9].

B. Clustered Federated Learning (CFL) and Data Heterogeneity

Clustered Federated Learning (CFL) extends the personalization concept by training separate cluster-specific models for groups of clients with similar data characteristics. By maintaining multiple models $\{W_1, W_2, \dots, W_K\}$, CFL effectively reduces Non-IID effects and improves convergence [10].

Existing CFL algorithms, such as IFCA and FedClust, employ implicit clustering, where cluster membership is dynamically determined by measuring similarity between local model updates or gradients [11]. However, these approaches require several initialization rounds and incur high communication overhead.

In contrast, CAP-FL adopts Explicit Feature-Based Clustering, where clients are pre-assigned to clusters based on known metadata—here, the play genre (Comedy, Tragedy, History). This strategy avoids iterative clustering and enables immediate, genre-specific model aggregation from the first communication round [12].

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Problem Formulation and Objective

CAP-FL adapts the standard Federated Averaging (FedAvg) algorithm [2] into a clustered and personalized optimization process. Let N denote the total number of clients divided into K disjoint clusters:

$$C = \{C_1, C_2, \dots, C_K\}, \quad C_i \cap C_j = \emptyset \text{ for } i \neq j \quad (1)$$

Each client $i \in C_k$ possesses its own dataset D_i . The global objective is to minimize the aggregated empirical loss across all clusters:

$$\min_{W_1, W_2, \dots, W_K} \sum_{k=1}^K \sum_{i \in C_k} L(W_k, D_i) \quad (2)$$

where $L(W_k, D_i)$ represents the cross-entropy loss for next-word prediction on local dataset D_i using the cluster-specific GPT-2 model W_k .

During local training, each client updates its parameters as:

$$W_{i,k}^{t+1} = W_k^t - \eta \nabla L(W_k^t; D_i) \quad (3)$$

B. Clustered Aggregation Mechanism

CAP-FL aggregates updates within each genre-based cluster:

$$W_k^{t+1} = \frac{\sum_{i \in C_k} |D_i| W_{i,k}^{t+1}}{\sum_{j \in C_k} |D_j|} \quad (4)$$

This selective aggregation minimizes inter-domain variance and accelerates convergence, producing a set of specialized

models $\{W_1, W_2, \dots, W_K\}$ that capture shared stylistic patterns within each genre.

C. Context-Aware Personalization Mechanisms

CAP-FL employs Context Token Injection (CTI). Each input sequence is embedded with contextual metadata:

$$X_{\text{context}} = \langle \text{SOS} \rangle + T_{\text{PLAY}} + T_{\text{GENRE}} + T_{\text{ACT}} + T_{\text{SCENE}} + T_{\text{CHARACTER}} + X_{\text{raw}} \quad (5)$$

where X_{raw} is the dialogue text, and T_* tokens provide explicit macro- and micro-context cues.

D. Local Fine-Tuning for Personalization

After cluster-specific aggregation, each client performs local fine-tuning:

$$W_i^* = W_k - \eta_p \nabla L(W_k; D_i) \quad (6)$$

ensuring that the client's model retains both global generalization and personal linguistic nuances.

E. System Implementation

The CAP-FL framework is implemented using the Flower Federated Learning library. Each Shakespeare play is treated as a separate client. Preprocessing includes normalization, token cleaning, context token generation, and sequence construction using a sliding window approach. Training is asynchronous to handle client dropouts, ensuring robust federated execution.

IV. EXPERIMENTAL SETUP

A. Environment and Libraries

The CAP-FL framework was developed and tested on a single workstation. The software environment consisted of:

- Python 3.8+
- PyTorch (CPU or CUDA if available)
- Hugging Face Transformers (distilgpt2)
- Flower (FLWR) for federated orchestration (simulation API)
- Standard Python libraries: pandas, numpy, os, and json

GPU utilization was automatically detected using: `torch.device("cuda" if torch.cuda.is_available() else "cpu")`.

To avoid repeated model downloads and possible network timeouts, all pretrained components were cached locally prior to large experiments (using `HF_HOME` or manual downloads).

B. Dataset and Preprocessing

The Shakespeare dataset was used, with each play representing an individual FL client. The preprocessing script (`split_client.py`) generated one JSONL file per play with entries formatted as:

```
{
  "input_sequence": "<SOS> <PLAY_xxx>
                    <GENRE_xxx>
                    <ACT...> <SCENE...>
```

```
<CHARACTER...> text",
  "target_word"   : "<EOS>",
  "play"          : "...",
  "genre"         : "..."
```

The preprocessing pipeline included:

- Token normalization and text cleaning.
- Context Token Injection (CTI): adding structured metadata tokens (`<PLAY...>`, `<GENRE...>`, `<ACT...>`, `<SCENE...>`, `<CHARACTER...>`).
- Sliding-window construction for input/target sequence pairs.

To accelerate experimentation, only the first 500 (configurable) sequences per client were used (`MAX_SEQUENCES_PER_CLIENT`), maintaining stylistic diversity while reducing computational overhead.

C. Model and Tokenizer

The base model used was DistilGPT-2—a distilled variant of GPT-2. Tokenizer configuration was performed as follows:

- Initialized using `AutoTokenizer.from_pretrained("distilgpt2")`.
- Added special tokens: `<PAD>`, `<SOS>`, `<EOS>`.
- Injected custom context tokens (e.g., `<GENRE_comedy>`) before resizing embeddings.

Embedding matrices were resized automatically to accommodate new tokens. Model utilities such as `get_gpt2_model()`, `ClientDataset`, and `evaluate_model()` were defined in `model_config_and_utils.py`.

D. Federated Setup and Hyperparameters

CAP-FL was executed as a clustered federated simulation reflecting the CAP-FL algorithm design. The configuration parameters are summarized below:

- Selected cluster: *Comedy*
- Total clients: 14 (Comedy plays)
- Clients per round: 4 (randomly selected)
- Federated rounds: 2
- Local epochs per client (cluster stage): 1
- Local epochs per client (personalization stage): 1
- Batch size: 4
- Learning rate: 5e-5
- Sequence length: 40 tokens
- Lines per client: 1000(configurable)

The Flower framework operated in *simulation mode*, allowing all virtual clients to run within a single process, mirroring realistic server–client coordination.

E. Training, Checkpointing, and Personalization

The complete training workflow (implemented in `fl_orchestrator.py`) proceeds as follows:

- 1) Initialize and save a global model (`global_model_wg_initial.pt`).

- 2) Create a cluster-specific model initialized from the global checkpoint.
- 3) For each federated round:
 - a) Randomly select four clients from the cluster.
 - b) Each client performs local fine-tuning (1 epoch).
 - c) The server aggregates client updates via FedAvg.
 - d) Evaluate and log cluster loss and perplexity.
- 4) Save the final cluster model (cluster_model_wgk_<cluster>_final.pt).
- 5) Perform client-level personalization by fine-tuning on each client's dataset, saving checkpoints (personalized_model_wi_<client>.pt).

All artifacts and logs are stored under data/models/. A results.json file records round-wise metrics and client selections.

Example results.json entry:

```
[
  {
    "round": 1,
    "selected_clients": ["merchantofvenice",
                        "tamingoftheshrew",
                        "loveslabourslost",
                        "amidsummernightsdream"],
    "avg_train_loss": 2.87,
    "eval_loss": 3.30,
    "perplexity": 30.16
  },
  {
    "round": 2,
    "selected_clients": ["asyoulikeit",
                        "tamingoftheshrew",
                        "twelfthnight",
                        "twogentlemenofverona"],
    "avg_train_loss": 1.67,
    "eval_loss": 1.99,
    "perplexity": 7.98
  }
]
```

F. Reproducibility and Runtime Optimizations

To ensure reproducibility and efficiency:

- Cache models locally (HF_HOME) to prevent repeated downloads.
- Enable GPU for substantial speedups.
- For rapid debugging, set MAX_SEQUENCES_PER_CLIENT = 100 to 500.
- When loading checkpoints, use map_location=DEVICE in torch.load() to prevent mismatches.
- For newer PyTorch versions, set weights_only=True for safer model loading.

V. EVALUATION METRICS

To evaluate the performance of the proposed Context-Aware Personalized Federated Learning (CAP-FL) framework, three

main quantitative metrics are utilized. These metrics assess predictive accuracy, contextual understanding, and overall model confidence.

A. Cross-Entropy Loss

Cross-Entropy Loss is the primary objective function in next-word prediction tasks [8]. It quantifies the divergence between the predicted probability distribution of the next word and the actual target distribution. A lower Cross-Entropy value indicates that the model's predicted probabilities align closely with the ground truth, reflecting improved convergence and learning stability [16]. In the CAP-FL framework, this loss is tracked at both cluster and client levels to assess global and personalized learning performance.

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{w \in V} y_{i,w} \log(p_{i,w}) \quad (7)$$

where $y_{i,w}$ represents the true word distribution, $p_{i,w}$ denotes the model's predicted probability, and V is the vocabulary set.

B. Accuracy

Accuracy measures the proportion of correct next-word predictions compared to the ground truth. While Cross-Entropy focuses on probability alignment, Accuracy provides an interpretable metric of prediction success [18]. In federated evaluation, accuracy is computed for each client and then averaged across genre clusters to analyze both personalization effectiveness and global generalization.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}} \quad (8)$$

Higher accuracy values demonstrate that the personalized CAP-FL models successfully adapt to user-specific linguistic patterns while maintaining syntactic and semantic coherence.

C. Perplexity (PPL)

Perplexity is a widely used metric in language modeling [19] to measure how well a probabilistic model predicts a given text. It is defined as the exponentiated average negative log-likelihood of the true words given the model's predictions:

$$\text{PPL} = \exp(L) \quad (9)$$

where L represents the average Cross-Entropy Loss. A lower perplexity value signifies stronger contextual comprehension and reduced uncertainty in next-word predictions. In CAP-FL, reduced perplexity across clients indicates the framework's ability to learn coherent, context-aware representations despite heterogeneous data distributions.

D. Evaluation Protocol

All metrics are computed for each client after both cluster-level aggregation and personalized fine-tuning. Results are averaged within each genre-based cluster (Comedy, Tragedy, History) to evaluate consistency and fairness of learning within similar linguistic groups. To maintain computational efficiency, evaluations are performed on a subset of 500–1000 dialogue lines per client [20], which adequately captures the model’s stability and adaptation behavior.

VI. RESULTS AND DISCUSSION

The proposed *Context-Aware Personalized Federated Learning* (CAP-FL) framework was implemented using DistilGPT-2 on Shakespeare plays. The Shakespeare dataset is a standard FL benchmark for next-word prediction [19], [20]. For demonstration, the *Comedy* genre was selected, which included 14 client datasets. To reduce computational load and simulate realistic federated training, 4 clients were randomly selected per round for model aggregation.

During federated cluster training, the global base model was initialized and saved successfully. Across two federated rounds, cluster models were updated by averaging the selected client weights in each round using the FedAvg algorithm [1]. After the second round, the final cluster model for the Comedy genre was saved, achieving an average loss of 1.9934 and perplexity of 7.98.

In the personalization phase, each client model was fine-tuned locally on its own dataset following the personalization scheme inspired by FedProx [2] and Contextualized FL [8]. The personalized models achieved significantly improved performance compared to the initial global model, with loss values ranging from 0.0796 to 0.6935 and perplexity ranging from 1.08 to 2.00. Table I summarizes the evaluation of the personalized models.

TABLE I
PERFORMANCE OF PERSONALIZED CAP-FL MODELS (COMEDY GENRE)

Client	Loss	Perplexity
acomedyoferrors	0.2095	1.23
allswellthatendswell	0.0827	1.09
amidsummernightsdream	0.0796	1.08
asyoulikeit	0.5022	1.65
awinterstale	0.3114	1.37
loveslabourslost	0.3562	1.43
measureformeasure	0.2518	1.29
merchantofvenice	0.5868	1.80
merrywivesofwindsor	0.1448	1.16
muchadoaboutnothing	0.5930	1.81
pericles	0.6935	2.00
tamingoftheshrew	0.2451	1.28
twelfthnight	0.6134	1.85
twogentlemenofverona	0.3174	1.37

The results demonstrate that CAP-FL successfully personalizes next-word prediction models for individual clients while maintaining contextual coherence within clusters. By limiting the number of clients per round, the system reduces training time without sacrificing personalization effectiveness.

The final personalized models are capable of generating text consistent with the style and genre of the respective plays.

Overall, the CAP-FL framework shows promising potential in enhancing user-specific text prediction performance in a federated learning environment while addressing computational constraints.

VII. CONCLUSION AND FUTURE WORK

The proposed Context-Aware Personalized Federated Learning (CAP-FL) framework effectively addresses the challenge of statistical heterogeneity in federated Next Word Prediction (NWP) tasks. By integrating Explicit Feature-Based Clustering and Context Token Injection (CTI), CAP-FL enables both macro- and micro-level personalization [8], [9]. This combination enhances contextual understanding and maintains stylistic coherence across diverse users.

Experimental evaluations on the Shakespeare Non-IID dataset demonstrate that CAP-FL achieves lower Cross-Entropy Loss and Perplexity, while attaining higher prediction accuracy compared to traditional federated learning methods such as FedAvg [1] and FedProx [2]. The framework achieves a balance between global collaboration and personalized adaptation, allowing models to learn unique linguistic patterns from users without compromising data privacy [10], [11].

The genre-based clustering promotes stable convergence by aggregating updates from stylistically similar clients [15], whereas CTI improves the model’s responsiveness to fine-grained contextual cues such as play, act, scene, and character. Together, these mechanisms enhance personalization, convergence stability, and fluency in generated text.

In summary, CAP-FL demonstrates that federated fine-tuning of transformer-based language models can provide efficient personalization and scalability even under highly Non-IID conditions [17]. The proposed system offers a privacy-preserving, interpretable, and communication-efficient solution for real-world predictive text applications.

A. Future Work

Future research directions include the following:

Parameter-Efficient Fine-Tuning (PEFT): Employing lightweight fine-tuning methods such as LoRA or QLoRA [3], [4], [6] to reduce communication overhead and memory requirements during local updates.

Dynamic Context Adaptation: Extending CTI to capture dynamic and user-specific contextual signals such as device activity, temporal cues, and topic context for real-time personalization.

Cross-Domain Validation: Applying CAP-FL to other natural language processing domains including email composition, conversational assistance, and code completion to evaluate its generalizability [7], [8].

Fairness and Scalability: Investigating techniques to ensure equitable model performance across users, balance cluster participation, and support large-scale deployment across heterogeneous edge devices [12]–[14].

Overall, CAP-FL lays a strong foundation for context-aware, privacy-focused personalization in Federated Language Modeling. Its dual-level adaptation strategy provides a path toward developing next-generation intelligent text systems that preserve user privacy while continuously learning from diverse language environments.

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